Patronising and Condescending Language Detection

NLP Coursework Report

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GitHub Repository	https://github.com/leonenkova/NLP_cw

For Exercise 4 and Exercise 5.1, please refer to the GitHub repository linked above.
BestModel/ contains the training notebook, dev.txt, and test.txt.

Exercise 1: Critical Paper Review

Q1. Primary Contributions

1. Novel dataset: The authors introduce a novel and publicly available dataset containing 10,637 news paragraphs collected from 20 English-speaking countries. These paragraphs focus on vulnerable communities and were selected using 10 keywords related to vulnerability. Each paragraph is labeled on a 5-point scale indicating the intensity of PCL. In addition, specific text spans that express PCL are carefully annotated, allowing for more detailed analysis at the phrase level.

2. A Two-Level PCL Taxonomy: To better understand how patronizing language appears in the media, the authors propose a linguistically informed taxonomy. It includes seven fine-grained categories — Unbalanced Power Relations, Shallow Solution, Presupposition, Authority Voice, Metaphor, Compassion, and The Poorer the Merrier. These are grouped under three broader archetypes: The Saviour, The Expert, and The Poet. This is the first taxonomy designed specifically to capture how PCL is used when discussing vulnerable communities in news media.

3. Baseline Experimental Analysis: The study also provides a comprehensive experimental evaluation across two main tasks: detecting whether a paragraph contains PCL (binary classification) and identifying the specific types of PCL present (multi-label classification). The authors compare traditional machine learning models (such as SVM with Bag-of-Words and word vectors), deep learning approaches (BiLSTM), and Transformer-based models (BERT, RoBERTa, and DistilBERT). These experiments establish strong baseline benchmarks, offering a solid foundation for future research in detecting and analyzing patronizing language.

Q2. Technical Strengths

1. Accurate Annotation Protocol: The dataset was created using a thoughtful and accurate annotation strategy. Three annotators reviewed each paragraph, and instead of assigning simple binary labels, the authors used a 5-point scale to capture borderline or ambiguous cases better. When strong disagreements occurred (such as a 0 vs. 2 rating), a referee annotator stepped in to resolve them. The paper also reports inter-annotator agreement (Kappa) for each category in Task 2, with scores ranging from 48% to 67%. This level of transparency and methodological care goes beyond what many dataset papers provide, where often only a single overall agreement score is reported.

2. Meaningful Error Analysis: The authors don't simply highlight where their models perform well — they also carefully examine where they fail. In Tables 5 and 6, they provide concrete examples of RoBERTa's mistakes, distinguishing between different types of errors. For instance, the model sometimes flags emotionally strong but non-patronizing language as PCL (false positives), while missing subtler cases such as The Poorer the Merrier (false negatives). This kind of detailed and honest analysis gives future researchers clear insight into remaining challenges and areas for improvement.

3. Balanced Geographical Coverage: The dataset includes news paragraphs from 20 English-speaking countries and is balanced across 10 vulnerability-related keywords. This deliberate sampling strategy helps reduce the geographic and topical bias that often appears in large web-scraped NLP datasets. As a result, the corpus better reflects the diversity of global English-language media.

Q3. Key Weaknesses

1. Severe Class Imbalance: Out of 10,637 paragraphs, only 995 (about 9.3%) are labeled as positive PCL cases. That means the dataset is heavily imbalanced. However, the paper does not clearly explain whether this imbalance was handled during training. While reporting F1-

score on the positive class in Task 1 helps somewhat, the authors don't examine how the imbalance may have affected model learning.

2. Moderate Inter-Annotator Agreement Raises Concerns: The paragraph-level inter-annotator agreement (IAA) is reported at 41% Kappa, which falls into the "moderate" range under the Landis & Koch scale. When borderline cases are excluded, agreement rises to 61%, suggesting that a substantial portion of the dataset involves uncertain or ambiguous labeling decisions. Although the authors acknowledge this issue, they don't provide additional validation to strengthen confidence in the final adjudicated labels. As a result, questions remain about how reliable the ground truth is.

3. Lack of Statistical Testing and Ablation Studies: The paper reports model performance using point estimates, but doesn't include confidence intervals or statistical significance testing. For example, the difference between RoBERTa (F1 = 70.63) and DistilBERT (F1 = 68.24) in Task 1 may not be statistically meaningful, especially given the use of 10-fold cross-validation. Without formal testing, it is unclear whether these differences reflect real improvements or random variation. Additionally, the paper doesn't include ablation studies to explain unexpected findings.

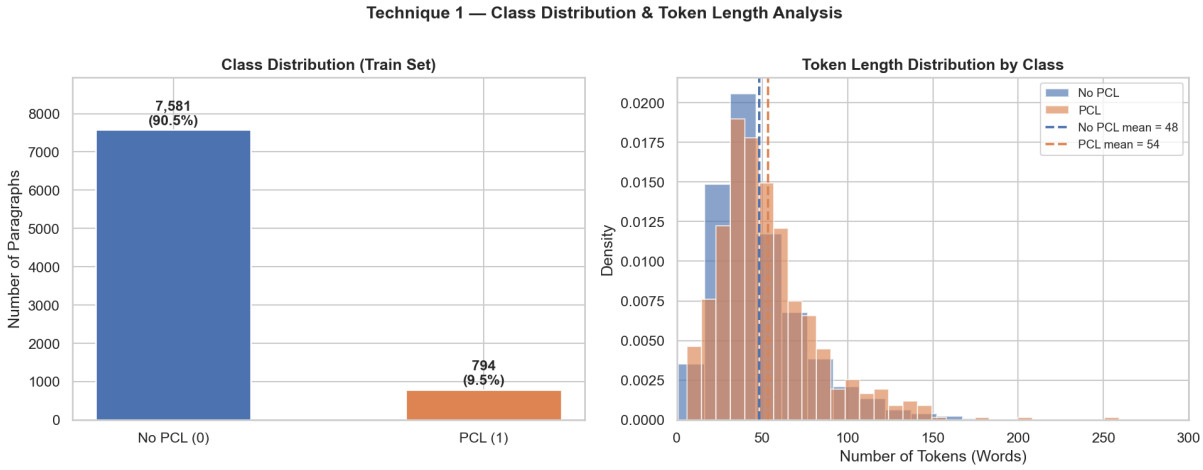
Exercise 2: Exploratory Data Analysis

Technique 1 — Class Distribution and Token Length

1a. Class Distribution

Class	Count	Proportion
No-PCL (label 0)	7,581	90.5%
PCL (label 1)	794	9.5%
Total	8,375	100%

1b. Token Length Distribution



Class	Mean tokens	Median tokens
No-PCL	48.2	42
PCL	54.1	46

The training dataset is highly imbalanced. About 90.5% of the paragraphs are labeled as **No PCL**, while only 9.5% are labeled as **PCL**, resulting in roughly a 9.5:1 ratio. This means a very simple model that always predicts "No PCL" would still achieve over 90% accuracy — despite completely failing to identify any actual PCL cases. This is precisely the kind of misleading outcome the appendix cautions against.

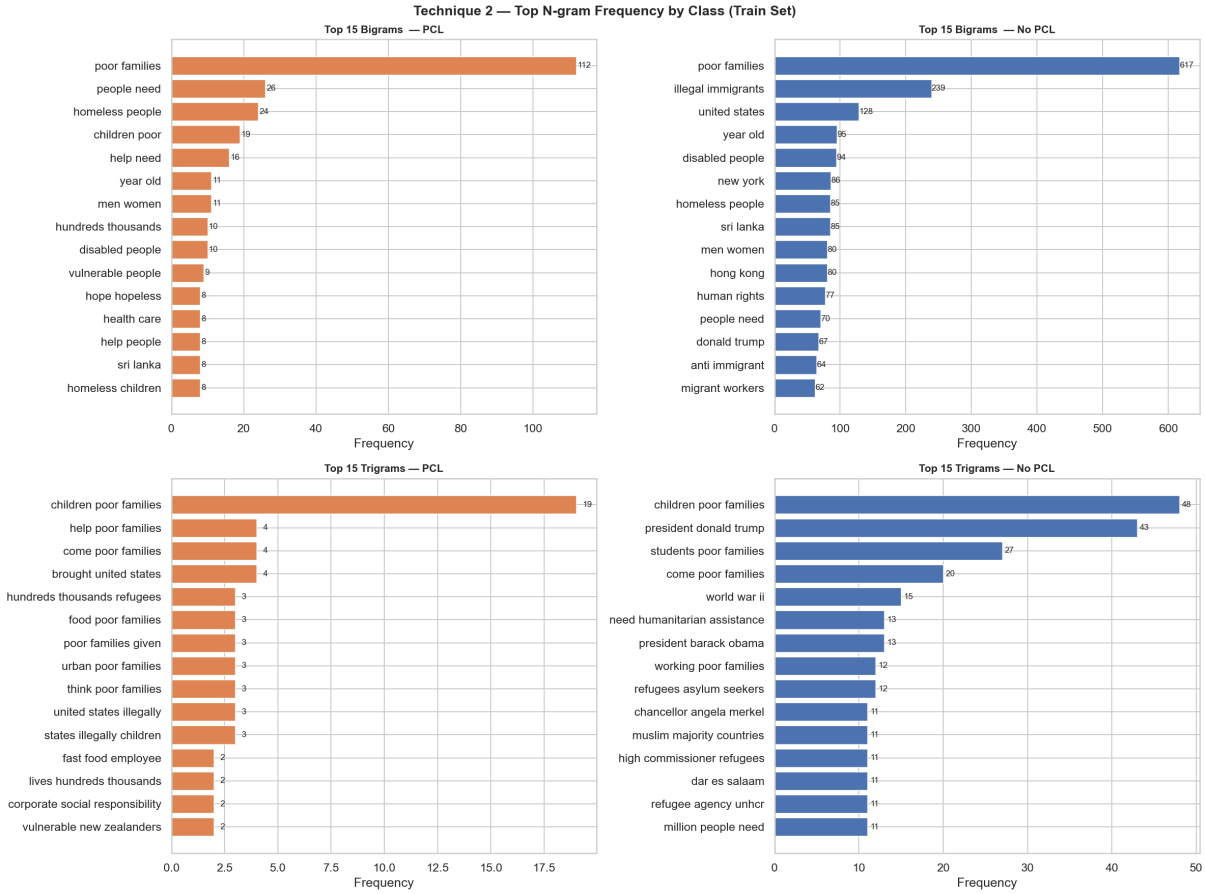
In terms of length, both classes follow a broadly similar distribution. However, PCL paragraphs are slightly longer on average (around 100 tokens) compared to No-PCL paragraphs (about 95 tokens). This suggests that patronizing or condescending language may be more likely to appear in somewhat longer, more elaborated passages. Importantly, very few paragraphs exceed 250 tokens, indicating that extremely long texts are rare in the dataset.

Overall Statements

1. The class imbalance needs to be explicitly addressed during training to prevent biased learning.
2. Setting max_length = 256 is appropriate and safely covers nearly all paragraphs.
3. Evaluating performance using F1-score on the positive class is the correct choice, as accuracy alone would be misleading in such an imbalanced setting.

Technique 2 — N-gram Frequency Analysis

Top-15 bigrams and trigrams were extracted separately for the PCL and No-PCL classes to identify discriminative lexical patterns.



Class	Top discriminative n-grams	Interpretation
PCL	"poor families", "people need", "help need", "those in need"	Clear framing of vulnerability and dependency — a strong signal of unbalanced power dynamics
No-PCL	"illegal immigrants", "united states", "prime minister"	Neutral, factual reporting focused on topics rather than tone

Looking more closely at the patterns, several linguistic trends stand out.

PCL-Specific Patterns

Certain bigrams — such as *"help them"*, *"in need"*, *"make a difference"*, and *"their lives"* — appear far more often in PCL paragraphs. These phrases reflect the *Saviour* framing described in the paper's taxonomy, where the writer or implied audience is positioned as the helper, and vulnerable groups are portrayed as passive recipients of aid.

Similarly, trigrams like *"help those in"* and *"change their lives"* reinforce this narrative. They closely align with categories such as *Unbalanced Power Relations* and *Shallow Solution***,** where complex social issues are simplified into narratives of rescue or transformation.

No-PCL Patterns

In contrast, No-PCL paragraphs are dominated by neutral journalistic expressions like *"said the"*, *"according to"*, and *"per cent."* These are typical markers of factual, objective reporting. They focus on delivering information rather than framing communities in a condescending or paternalistic way.

Overall Statements

1. Keyword-based features can help identify patterns, but they are not sufficient on their own.
2. Models should avoid relying too heavily on surface-level lexical cues, as this may lead to shallow pattern matching rather than genuine understanding.
3. Data augmentation strategies should be carefully designed to preserve meaningful PCL signal phrases instead of accidentally removing the very linguistic markers that define the phenomenon.

Exercise 3: Proposed Approach

The official baseline is a fine-tuned RoBERTa-base model achieving F1 = 0.48 on the official development set. Exploratory data analysis revealed the most critical issue in the dataset: a severe class imbalance of approximately 9.5:1 (No-PCL : PCL).

To address this imbalance and optimise performance for the task metric (F1 on the positive class), the proposed approach introduces three targeted modifications:

1. Class-weighted loss
2. Decision threshold optimisation
3. Adaptive learning rate scheduling

Step 1 — Class-Weighted Cross-Entropy Loss

Instead of standard cross-entropy (which assigns equal importance to all examples), I used a class-weighted cross-entropy loss to cope with imbalance.

Class weights are computed using the inverse-frequency formulation:

$$w_k = \frac{N}{2 * N_k}$$

where N = 8,375 (total training examples), N_0 =7,577 (No-PCL examples), N_1 =794 (PCL examples). This yields: w_0 = 0.552 (No-PCL), w_1 =5.27 (PCL)

To avoid aggressive reweighting that can harm calibration and training stability, I applied a softening scale factor of 0.7:

$$w_k = \frac{N}{2 * N_k} * 0.7$$

Class	N_k	Weight	Effect
No-PCL (k=0)	7,581	0.387	Down-weighted — common class
PCL (k=1)	794	3.69	Up-weighted — rare class

Thus, misclassifying a PCL sample is penalised approximately 9.5x more than misclassifying a No-PCL sample, matching the empirical imbalance ratio.

Additionally, I applied label smoothing (ε = 0.1) to reduce overconfident predictions, improve probability calibration and stabilise training under heavy reweighting. Label smoothing distributes 10% probability mass uniformly across classes, preventing extreme logits and improving threshold stability.

Step 2 — Adaptive Learning Rate

Training uses a two-phase learning rate schedule. First, a manual linear warmup ramps the LR from 0 to 1×10^{-5} over 15% of total training steps, preventing large gradient updates on an untrained classification head. After warmup, ReduceLROnPlateau monitors dev F1 at the end of each epoch and halves the LR (factor=0.5) if no improvement is seen for 1 epoch, down to a minimum of 1×10^{-7}. This removes the need for manual LR tuning and enables fine-grained convergence in later epochs.

Step 3 — Decision Threshold Optimisation

After training, the model outputs a probability score $p \in [0,1]$ for each paragraph. The default threshold of 0.5 (predict PCL if p > 0.5) is not optimal for imbalanced tasks where F1 score on the positive class is the target metric. I performed a grid search over thresholds $t \in \{0.1, 0.15, \dots, 0.9\}$ on the official dev set, selecting:

$$t^* = argmax\ F1(PCL)\ over\ t \in [0.1, 0.9]$$

The optimal threshold t^* is then applied to the test set predictions. This step requires no additional training and adds negligible compute cost.

Training Hyperparameters

Hyperparameter	Value	Justification
Base model	microsoft/deberta-v3-base	Strong contextual encoder; competitive baseline
Batch size	16	GPU memory vs. gradient noise trade-off
Learning rate	1e-5	Standard for DeBERTa fine-tuning
LR schedule	Manual linear warmup 15% + ReduceLROnPlateau	factor=0.5, patience=1; prevents large early updates and adapts LR on plateau
Epochs	6 (early stop)	Patience=3 on dev F1 to prevent overfit
Loss	CrossEntropyLoss	w_neg=0.387, w_pos=3.69, scale=0.7, label_smoothing=0.1
Threshold	t* = 0.50 (well-calibrated via label smoothing)	argmax F1_PCL over [0.1, 0.9]

Exercise 5: Evaluation

Exercise 5.2 — Local Evaluation

Model	Dev F1 (PCL)	Test F1 (PCL)	vs. Baseline (dev)
Baseline	0.4800	0.4900 (est.)	—
DeBERTa-v3-base	0.6043	0.5497*	+0.1243 (+25.9%)

* *Test F1 estimated locally against publicly released ground-truth labels.*

The proposed model outperforms the baseline. On the development set, F1 improves from 0.48 to 0.6043 — a relative gain of nearly 26%. The estimated test F1 of 0.5497 confirms that the improvement generalizes beyond the dev split.

Error Analysis

Confusion Matrix

	Predicted No-PCL (0)	Predicted PCL (1)
Actual No-PCL (0)	1,833 (TN)	62 (FP)
Actual PCL (1)	86 (FN)	113 (TP)

Metric	Value
Precision (PCL)	0.6457
Recall (PCL)	0.5678
F1 (PCL)	0.6043
Accuracy	0.9293

The model demonstrates a balanced but precision-leaning behaviour. Precision 0.6457 indicates that when the model predicts PCL, it is correct roughly 65% of the time. Recall 0.5678 shows that the model detects about 57% of all true PCL instances. Since precision is higher than recall, the model is slightly more conservative in predicting the minority class. In other words, it avoids false positives more than it prioritises capturing every PCL case.

False Positive Analysis

All 62 FPs have original labels {0, 1}, meaning they are officially non-PCL or borderline cases. The dominant pattern is surface keyword triggering: the model fires on any text mentioning vulnerable groups, confusing "topic" with "tone".

Example text (truncated)	Keyword	Orig label
"The Jali family... donated wheelchairs to their disabled members."	disabled	0
"Marcos said the government should help poor families that try every possible means to survive."	poor-families	1
"So we do need to heal ourselves as an Aboriginal Torres Strait islander community..."	hopeless	0

The model has partially learned correlations between vulnerable-group topics and PCL labels. It has not fully disentangled reporting about vulnerable groups from talking down to them.

False Negative Analysis

All 84 false negatives come from strong PCL labels {2, 3, 4} with high annotator agreement. These cases are more subtle. The condescension is not expressed through obvious vocabulary but through framing, presupposition, or narrative positioning.

Example text (truncated)	Keyword	Orig label
"His present 'chambers' may be quite humble, but Shiyani has the tiny space very neatly organized..."	homeless	3
"Sheepherding in America has always been an immigrant's job, too dirty, too cold and too lonely for anyone with options."	immigrant	3
"...leave many of the most vulnerable people..."	poor-families	4

The model struggles with pragmatic implicature. Many false negatives require understanding what is implied rather than explicitly stated. A sentence-level encoder fine-tuned on limited data may not reliably capture these deeper framing effects.